Introduction to Machine Learning

Lecture Notes for Undergraduate Students (Test Document)

1. What is Machine Learning?

Machine learning is a branch of artificial intelligence in which computers learn patterns from data instead of following hand-written rules. The term was coined by Arthur Samuel in 1959 while he worked on a checkers-playing program that improved with experience.

There are three main families of machine learning: supervised learning, unsupervised learning and reinforcement learning. They differ mainly in the kind of feedback the algorithm receives while it learns.

2. Supervised Learning

In supervised learning every training example comes with a label. Classification predicts a category, such as whether an email is spam, while regression predicts a number, such as the price of a house.

Common algorithms include linear regression, decision trees and support vector machines. A typical practice is to split the data into 80 percent for training and 20 percent for testing, so the model is judged on examples it has never seen.

3. Unsupervised Learning

Unsupervised learning works with data that has no labels. Its goal is to discover hidden structure. Clustering groups similar items together; the K-means algorithm, for example, assigns each point to one of k clusters by minimising the distance to the cluster centre.

Dimensionality reduction methods such as principal component analysis (PCA) compress many features into a few. A common business use is customer segmentation, where shoppers with similar habits are grouped without anyone defining the groups in advance.

4. Overfitting and Underfitting

Overfitting happens when a model memorises the training data, including its noise, and then performs poorly on new data. Underfitting is the opposite problem: the model is too simple to capture the real pattern.

Remedies for overfitting include collecting more data, applying regularisation, using early stopping and validating with cross-validation. In k-fold cross-validation the data is divided into k parts, and k is commonly set to 5 or 10.

5. Evaluating Models

Accuracy is the share of predictions that are correct, but it can be misleading on imbalanced data. If 99 percent of transactions are legitimate, a model that always predicts legitimate is 99 percent accurate yet useless for catching fraud.

Precision measures how many predicted positives are truly positive, and recall measures how many true positives were found. The F1 score is the harmonic mean of precision and recall. A confusion matrix shows true positives, false positives, true negatives and false negatives in one table.

6. Neural Networks

A neural network is made of layers of connected neurons, each with adjustable weights. Training uses backpropagation to compute how each weight contributed to the error, and gradient descent to adjust the weights step by step.

The learning rate controls the size of each step: too large and training becomes unstable, too small and it is very slow. The ReLU activation function is popular because it is simple and helps avoid vanishing gradients. Deep learning simply means using networks with many layers.

7. Transformers and Modern NLP

The attention mechanism let models weigh the importance of every word in a sentence against every other word. The Transformer architecture, introduced in 2017 in the paper titled Attention Is All You Need, is built entirely on attention.

BERT, released in 2018, reads text in both directions and produces rich embeddings that capture meaning. Retrieval-augmented generation (RAG) combines such embeddings for search with a language model that writes the final answer, which reduces hallucination.